

Aayush Gadamshetty

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EDUCATION

Texas A&M University

College Station, TX

Bachelor of Science in Computer Science, Minor in Cybersecurity, GPA: 3.5

May 2027

TECHNICAL SKILLS

Languages & Databases: Python, SQL, Java, C++, Go, R, Linux/Bash, PostgreSQL, Distributed SQL

Big Data & Cloud: PySpark, Databricks, Distributed Computing, ETL Pipelines, AWS (EC2, S3, Lambda)

ML & Practices: Feature Engineering, Data Modeling, UMAP, t-SNE, Pandas, NumPy, Scikit-Learn, Statistical Modeling Data Integrity/Validation, Monitoring & Alerting, Git/GitHub, Agile/Scrum, Tableau/Dashboards

EXPERIENCE

Undergraduate Research Assistant (Big Data & ETL Pipelines)

Aug. 2025 – Aug. 2026

Texas A&M University

College Station, TX

- Architected scalable **Big Data pipelines and ETL workflows** in **Python** and **PySpark** to ingest, clean, and process high-dimensional datasets of 5,000+ records.
- Engineered distributed feature extraction scripts utilizing **UMAP** and **t-SNE**, optimizing data preprocessing overhead by 25% to support automated analytics and ML clustering models.
- Enforced rigorous data validation, data integrity protocols, and automated data wrangling to ensure reproducibility across multidisciplinary research teams.

Backend Development Specialist (Data Integration & APIs)

Sep. 2024 – Jun. 2025

Quila

Dallas, TX

- Developed resilient data pipelines, microservices, and workflow automation systems using **Python** and **AWS** infrastructure to support AI Assistant and analytics modules.
- Maintained distributed APIs and implemented data observability, logging, and performance monitoring to identify system bottlenecks and improve data retrieval speeds.
- Collaborated within an **Agile Scrum** framework to design scalable data models and automate ETL handoffs from MVP prototyping to production pipelines.

Systems Support & IT Operations Intern

Jun. 2023 – Oct. 2023

Quick MD Care

Frisco, TX

- Administered secure on-premises and cloud IT infrastructure, handling access control and ensuring 99.9% uptime for critical organizational software.

PROJECTS & OPEN SOURCE CONTRIBUTIONS

Open Source Contributor (Data Pipelines & Enterprise APIs) | Python, Go

Dec. 2025 – Jun. 2026

- **Google Cloud Python & NVIDIA DALI:** Explored and contributed to enterprise-grade APIs and **GPU-accelerated data loading pipelines** for machine learning models, navigating massive codebases.
- **NATS Server:** Contributed to the high-performance open-source messaging server codebase, navigating complex distributed systems architecture and low-latency data streaming.

Automated Trading & Statistical Modeling Engine | Python, AWS, ETL

Sept. 2024 – Jan. 2025

- Engineered an automated low-latency data ingestion pipeline in **Python** and **AWS** to extract, transform, and load historical market signals for downstream statistical modeling.
- Executed quantitative feature engineering and risk modeling scripts, outperforming benchmarks by 4.52% while achieving a Sharpe ratio of 0.952 and a 1.824 profit factor.

Scalable POS System & Analytics Dashboard | React, TypeScript, PostgreSQL

Jan. 2026 – May 2026

- Engineered an actionable full-stack web application and dashboard, integrating a distributed **PostgreSQL** database optimized for fast data retrieval and reporting.
- Developed automated data seeding scripts and managed database administration across a 5-person engineering team utilizing strict Agile/Scrum sprint methodologies.

COURSEWORK

Core Coursework: Parallel Computing (CS 435), Machine Learning (CS 421), Artificial Intelligence (CS 421), Analysis of Algorithms (CS 411), Operating Systems (CS 410), Computer Systems (CS 313), Data Structures (CS 221)